Invariant Graph Learning via Environment Augmentation Learn Invariance?

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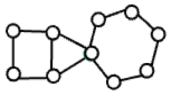
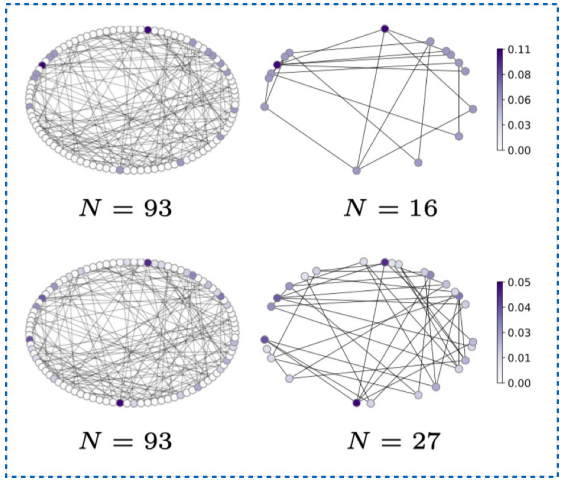
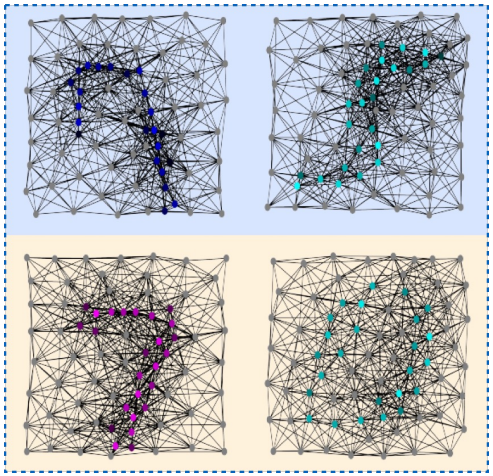
Out-of-Distribution Generalization on Graphs

OOD generalization on graphs is fundamentally more **challenging** than that on Euclidean data:

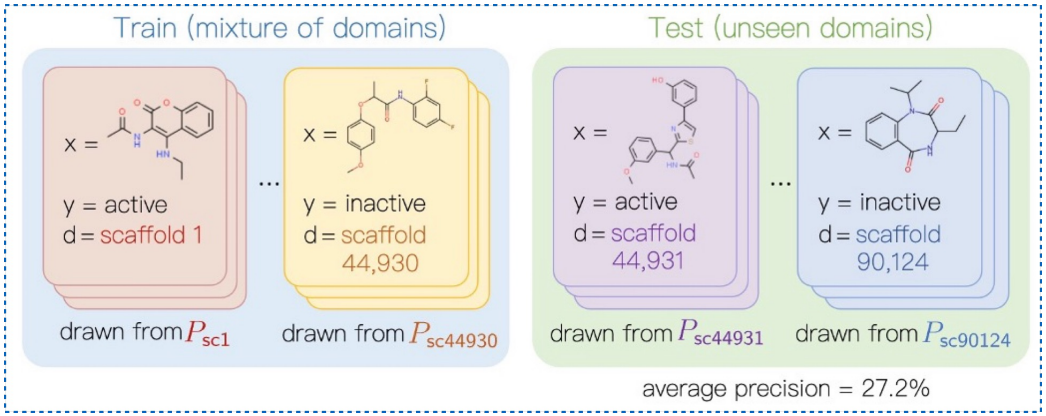
f_{GNN}({  } , { }) = **“House”**



Structure-level shifts



Attribute-level shifts



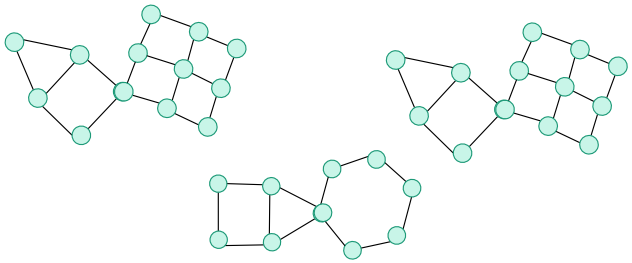
Mixture of structure-level and attribute-level shifts

(Knyazev et al. 2019; Hu et al., 2020; Koh et al., 2021; Gui et al., 2022; Chen et al., 2022)

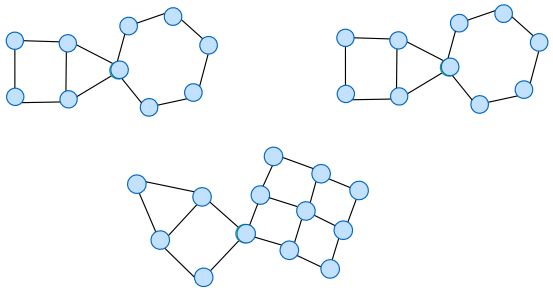
Out-of-Distribution Generalization on Graphs

Invariant graph representation learning aims to identify an **invariant subgraph** among graphs from different **environments** or **domains**:

Environment #1: Class “House”



Environment #2: Class “House”



(Wu et al., 2022ab; Miao et al., 2022; Chen et al., 2022)

Out-of-Distribution Generalization on Graphs

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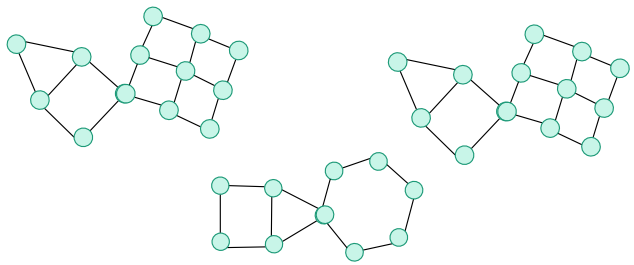
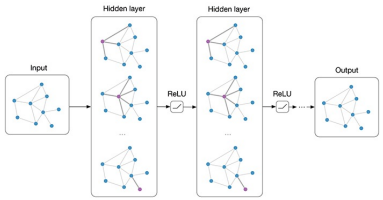
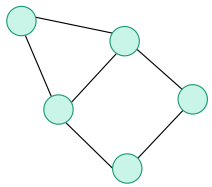
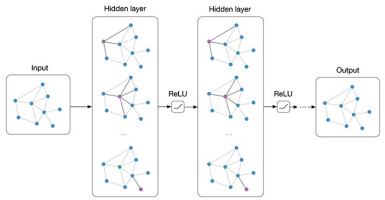




Extractor

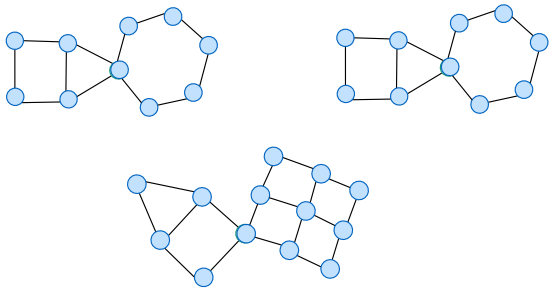




Classifier

“House”

Environment #2: Class “House”



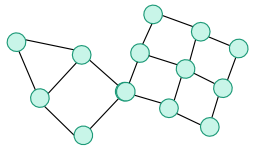
Extracted Invariant Subgraph

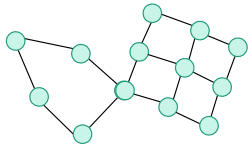
(Wu et al., 2022ab; Miao et al., 2022; Chen et al., 2022)

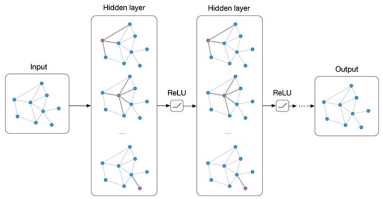
The “Free Lunch Dilemma” in OOD Generalization on Graphs

However, the **environments** or **domains**: information are usually expensive to obtain for graph structured data:

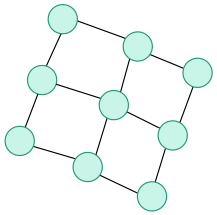
Environment #?: Class “House”

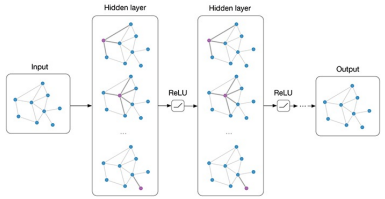






Extractor





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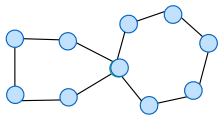
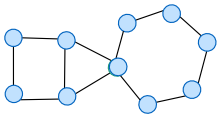
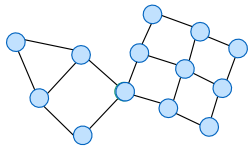






Extracted “Invariant” Subgraph

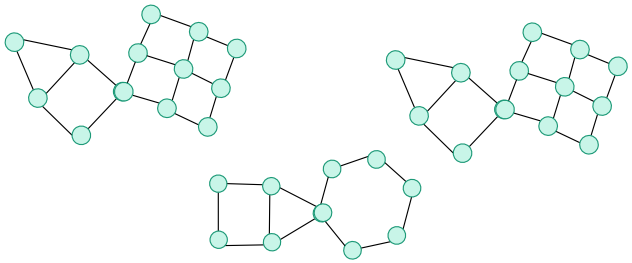
(Wu et al., 2022ab; Miao et al., 2022; Chen et al., 2022)

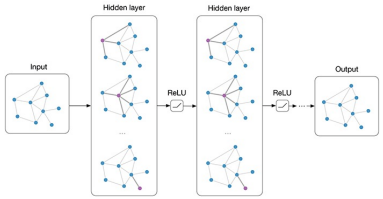
Is it possible to augment the environment information to enable OOD generalization on graphs?

The “Free Lunch Dilemma” in OOD Generalization on Graphs

One line of works aim to generate **new environments** based on the existing extracted subgraphs:

Environment #1: Class “House”





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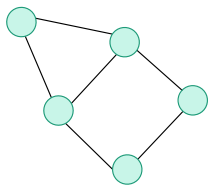
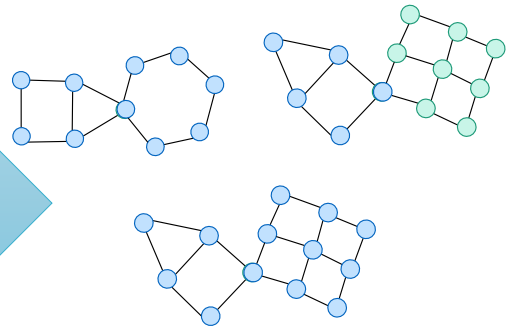


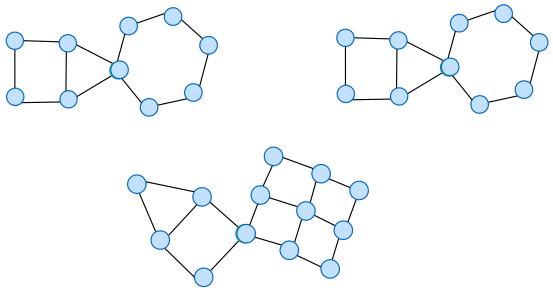
Extracted “Invariant” Subgraph

Generator

Environment #3: Class “House”



Environment #2: Class “House”



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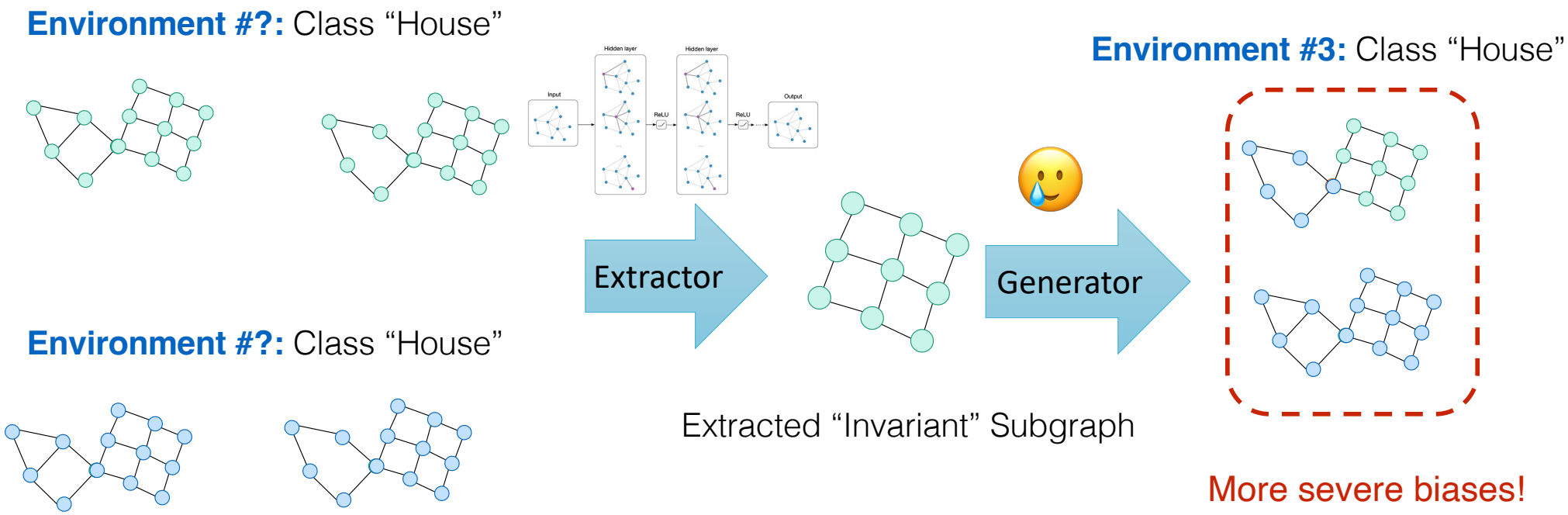

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The “Free Lunch Dilemma” in OOD Generalization on Graphs

Another line of works aim to **infer environment labels** for learning the underlying invariance:

Environment #?: Class “House”

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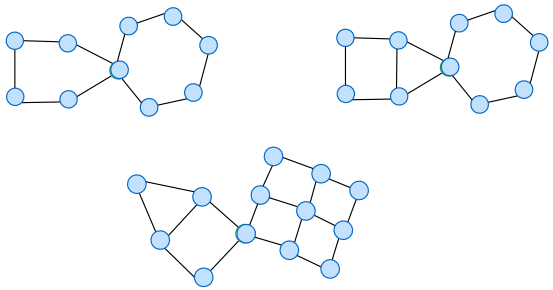
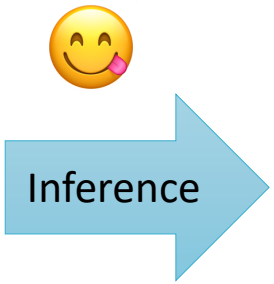
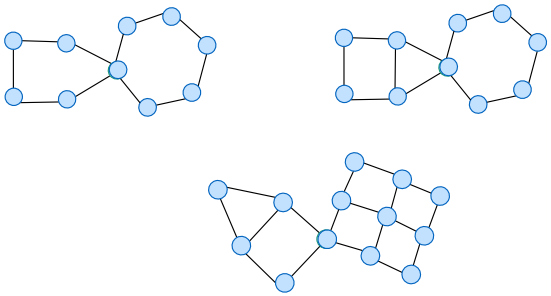




Environment #1: Class “House”

Environment #2: Class “House”



(Li et al., 2022; Yang et al., 2022)

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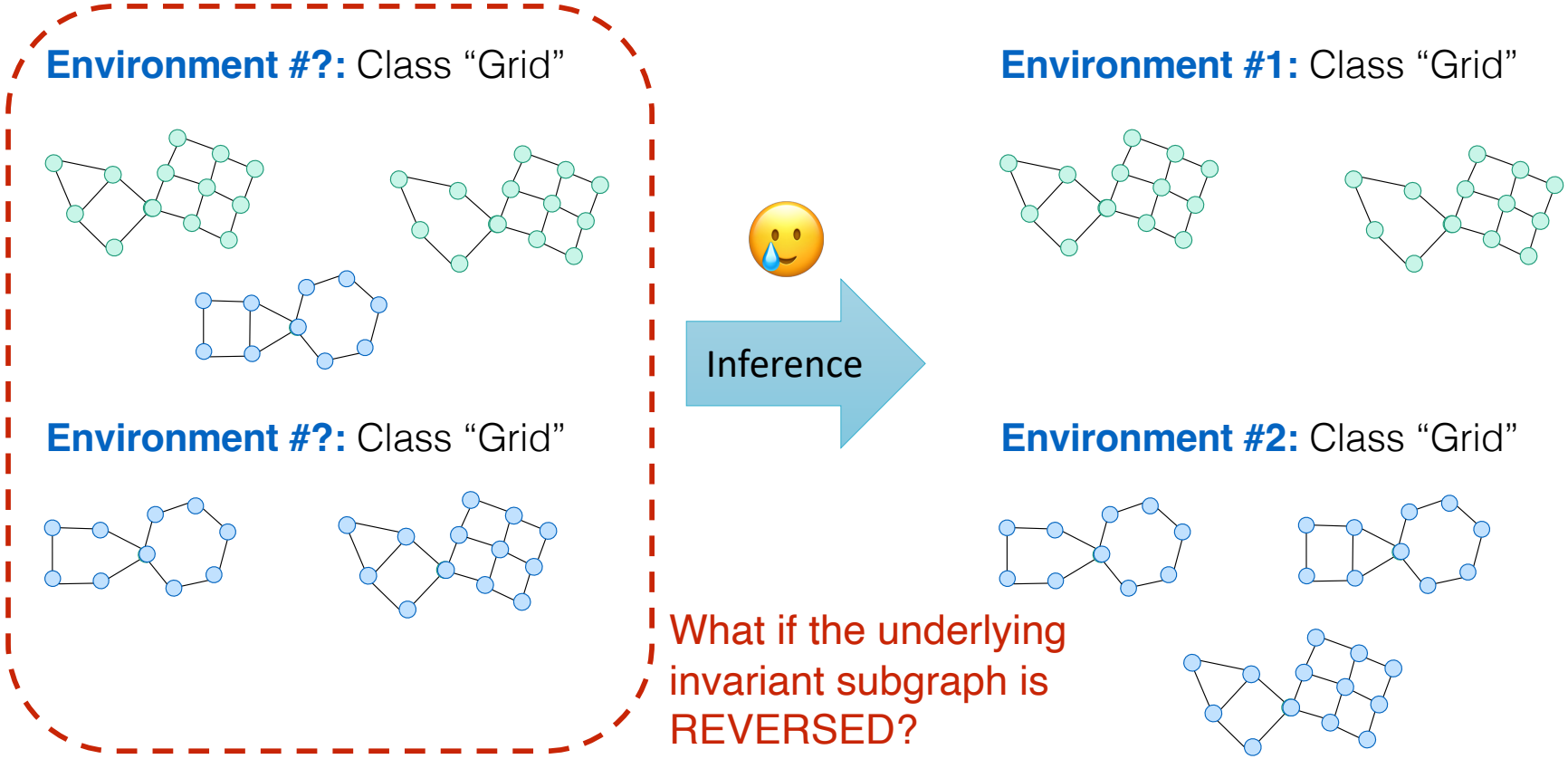
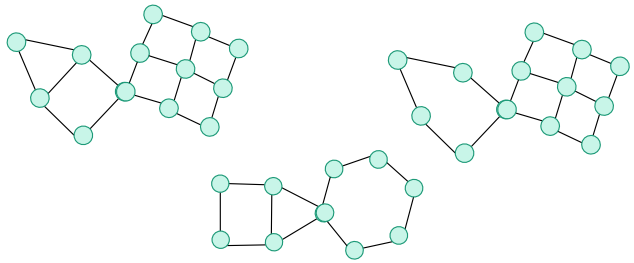


(Li et al., 2022; Yang et al., 2022)

Impossibility Results for OOD Generalization on Graphs

OOD generalization on graphs is fundamentally more **challenging** than that on Euclidean data:

Environment #?: Class “House”



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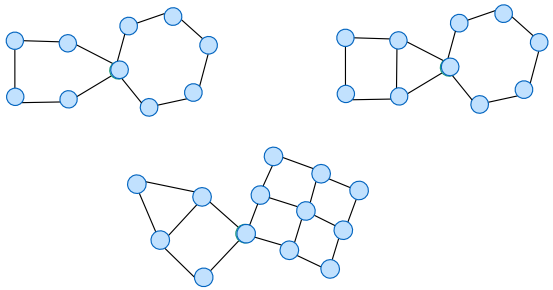
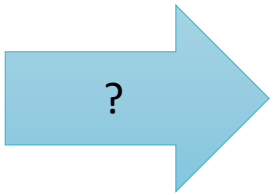




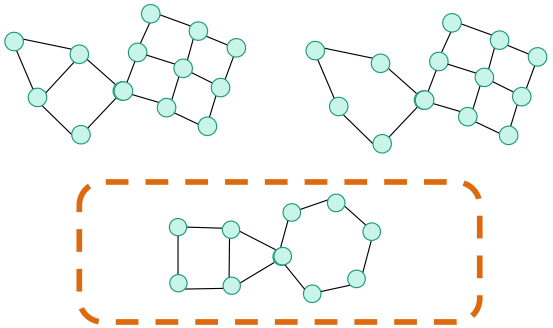
No Free Lunch in Graph OOD (Informal)
It is fundamentally ***impossible*** to identify the underlying invariant subgraph without further inductive biases.

What are the minimal sufficient inductive biases for invariant graph representation learning?

Failures of Environment Generation

How can we address **environments generation** failures?

Environment #?: Class "House"



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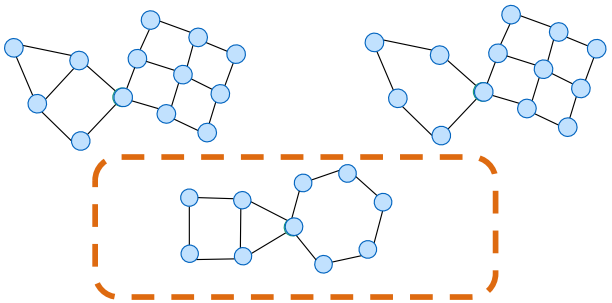
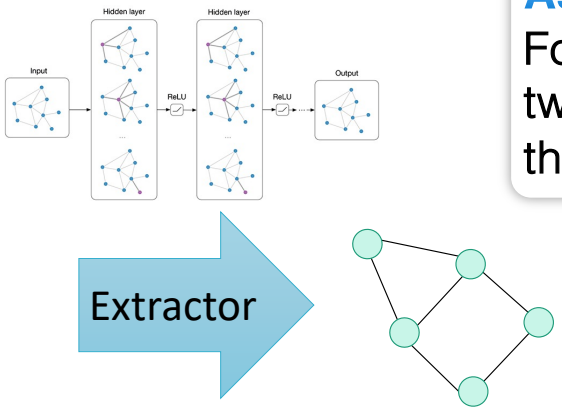




Assumption 1 (Variation Sufficiency)
For any spurious subgraph, there exists two underlying environments, such that the spurious correlation varies.

Extracted Invariant Subgraph

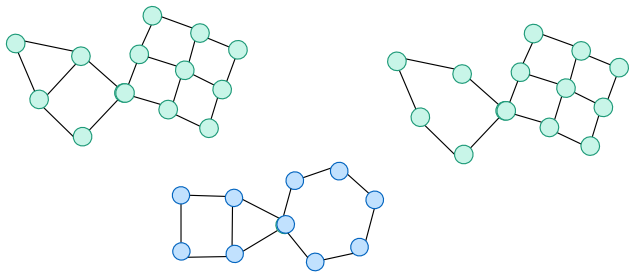
(Wu et al., 2022ab; Liu et al., 2022)

Failures of Environment Inference

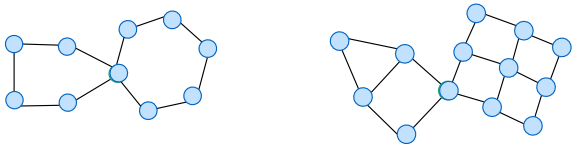
How can we address **environment inference** failures?

Either

Environment #?: Class "House"



Environment #?: Class "House"



OR

Environment #?: Class "Grid"

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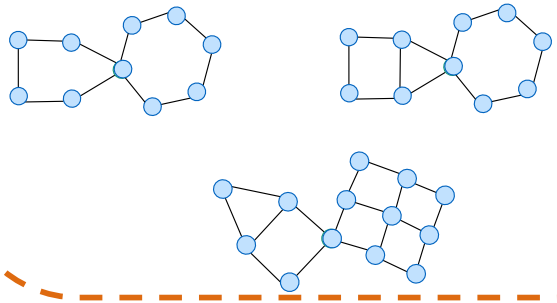
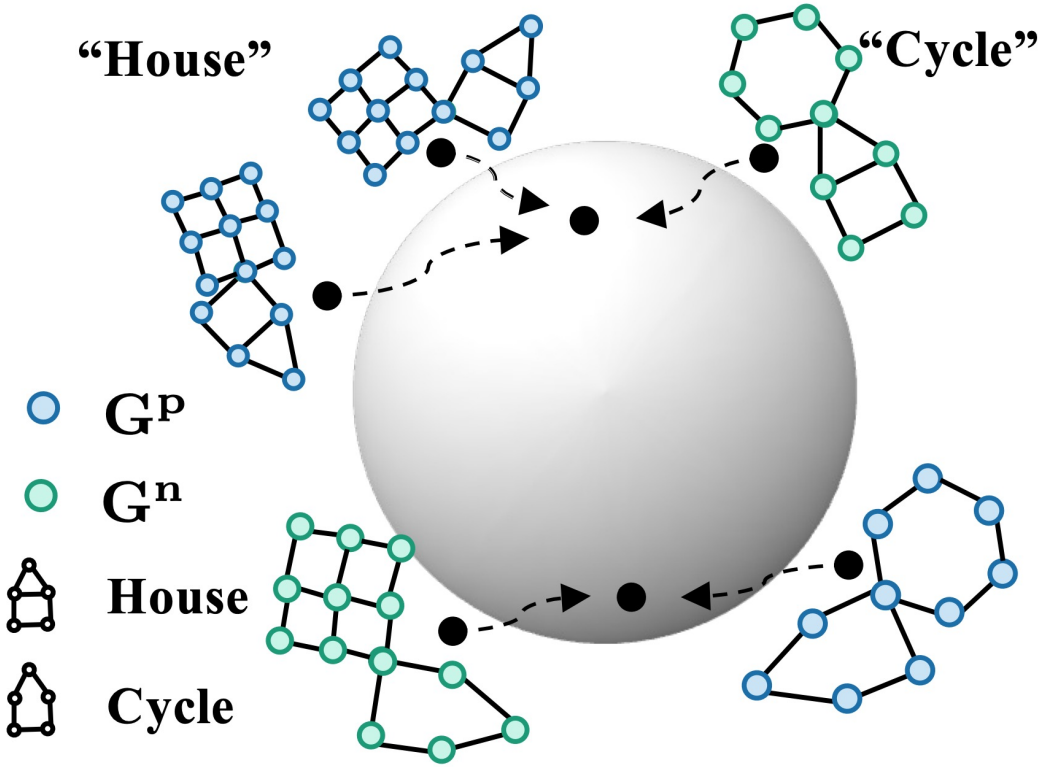


Assumption 2 (Variation Consistency)
For all environments, either spurious correlation is stronger or weaker.

(Li et al., 2022; Yang et al., 2022)

GALA: invariant GrAph Learning Assistant

Improving the contrastive invariant subgraph extraction via an **Environment Assistant**:



Experimental Results

GALA consistently improves the OOD generalization performance under various graph distribution shifts:

Table 1: OOD generalization performance under realistic graph distribution shifts.

Datasets	EC50-Assay	EC50-Sca	EC50-Size	CMNIST-sp	Graph-SST2	Graph-SST5	Twitter	Avg (Rank)†
ERM	76.42 (1.59)	64.56 (1.25)	62.79 (1.15)	21.56 (5.38)	81.54 (1.13)	42.62 (2.54)	59.34 (1.13)	58.40 (7.14)
IRM	76.51 (1.89)	64.82 (0.55)	63.23 (0.56)	23.29 (7.82)	82.22 (0.88)	42.77 (1.26)	60.42 (1.06)	59.04 (5.29)
V-Rex	76.73 (2.26)	62.83 (1.20)	59.27 (1.65)	24.62 (8.75)	79.27 (1.95)	42.48 (1.67)	60.50 (2.05)	57.96 (7.57)
EIIL	76.96 (0.25)	64.95 (1.12)	62.65 (1.88)	24.55 (13.3)	80.70 (1.21)	43.79 (1.19)	60.15 (1.44)	59.11 (5.00)
IB-IRM	76.72 (0.98)	64.43 (0.85)	64.10 (0.61)	13.06 (1.97)	82.12 (1.43)	43.02 (1.94)	60.80 (2.50)	57.75 (5.57)
GREA	73.47 (2.48)	62.00 (1.36)	61.88 (1.00)	18.64 (6.44)	80.20 (0.56)	43.29 (0.85)	59.92 (1.48)	57.06 (8.57)
GSAT	66.87 (8.70)	63.31 (1.17)	61.70 (1.48)	12.77 (2.00)	80.55 (2.18)	43.24 (0.61)	60.13 (1.51)	55.51 (8.57)
DisC	65.40 (5.34)	54.97 (3.86)	56.79 (2.56)	54.07 (15.3)	79.10 (2.09)	40.67 (1.19)	57.89 (2.02)	58.41 (10.1)
MoleOOD	61.94 (1.90)	59.53 (3.47)	56.08 (1.45)	39.55 (4.35)	80.78 (1.13)	40.36 (1.85)	59.26 (1.67)	56.79 (9.71)
GIL	72.13 (4.70)	63.05 (1.04)	62.08 (1.60)	18.04 (4.39)	82.04 (0.97)	43.30 (1.24)	61.78 (1.66)	57.49 (6.29)
CIGAv1	78.46 (0.45)	**66.05 (1.29)**	65.35 (0.88)	23.66 (8.65)	81.97 (0.87)	44.05 (1.48)	61.15 (0.72)	60.10 (3.00)
GALA	**79.24 (1.36)**	66.00 (1.86)	**66.01 (0.84)**	**59.16 (3.64)**	**82.50 (0.86)**	**44.88 (1.02)**	**62.45 (0.62)**	**65.75 (1.14)**
Oracle (IID)	85.18 (1.13)	83.02 (0.77)	86.07 (0.33)	67.25 (0.76)	91.40 (0.26)	47.96 (1.34)	63.66 (0.79)	

† Averaged rank is also reported in the parentheses because of dataset heterogeneity. Lower rank is better.

Summary

We conducted a retrospective study on the faithfulness of the augmented environment information for OOD generalization on graphs.

By showing the impossibility results, we developed a set of minimal assumptions for feasible invariant graph learning.

we proposed a provable feasible approach GALA under the assumptions. Extensive experiments with 11 datasets verified the superiority of GALA.

Paper

Thank you!

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